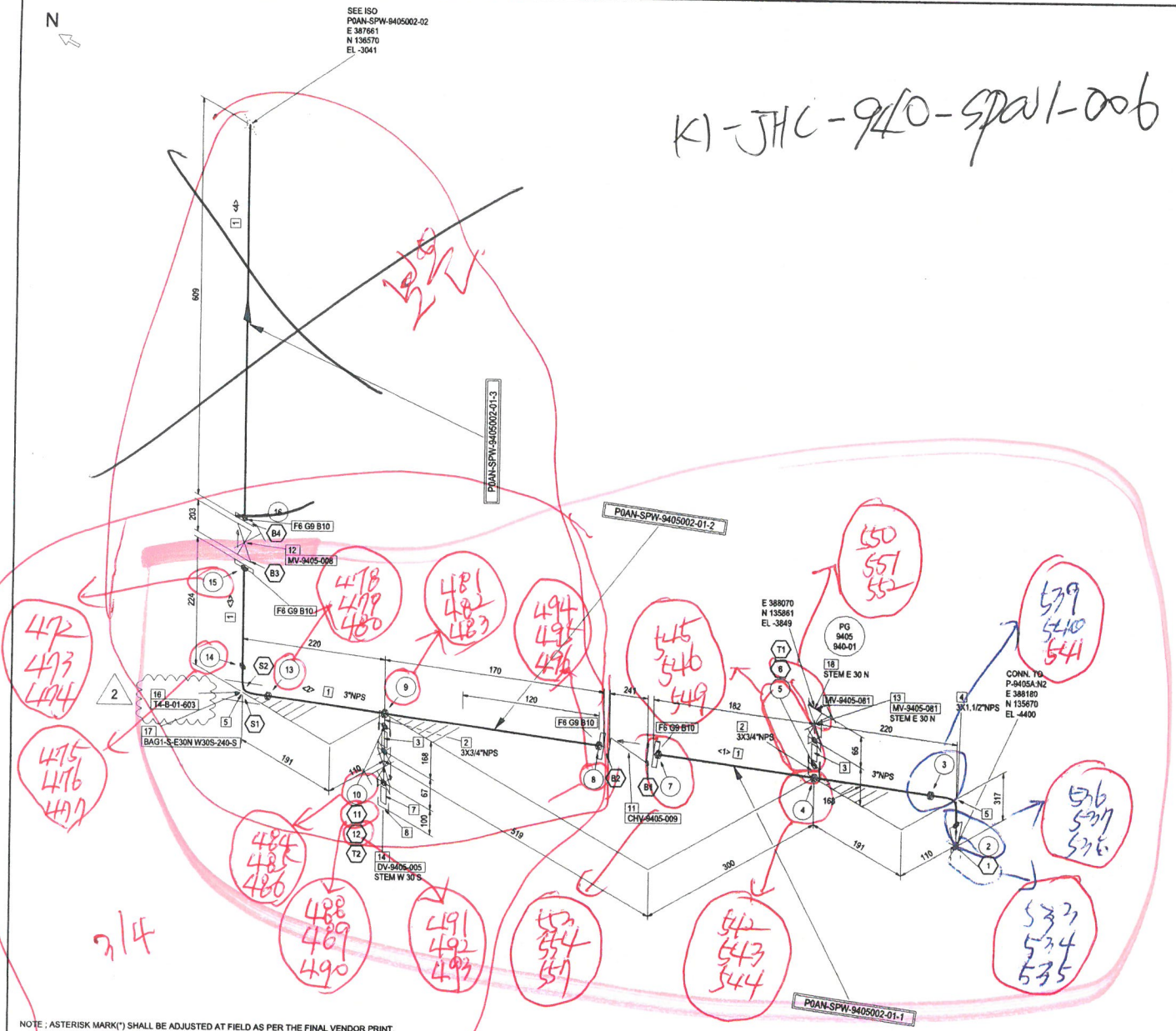

K1-JHC-940-SP001-006

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
1.5	S98	2.7	80	5	128	-	S	ISN	25		N	5	K1-103
3	S98	2.7	80	5	128	-	S	ISN	25		N	5	K1-103
0.75	S98	2.7	80	5	128	-	S	ISN	25		N	5	K1-103

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2	26-12-25	ISSUED FOR REVISION	N.H.PARK	M.H.SUNG	T.H.LIM
1	14-06-25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	15-04-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	3"	1.3 M	S98	P	PA02-000001903
2	HALF COUPLING SW CL3000 ASTM A182-F316/316L	3X3/4"	2	S98	TH	PB99-000038951
3	PIPE NIPPLE PBE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	2	S98	NPPG	PB99-000044539
4	CON. SWAGE BLE/PSE SMLS ASTM A403-WP316/316L SCH10S * SCH40S	3X1.1/2"	1	S98	RCS	PB99-000102550
5	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	3"	2	S98	E	PB99-000038410
6	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	3"	4	S98	F	PB01-000023956

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
7	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
8	CAP SCR'D (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
9	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK	3"	4	S98	G	PF01-000009268
10	100 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UN/U NR x MM	5/8"	16	S98	B	PB99-000076890
11	CHECK VALVE FLGD-RF CL 150 ASTM A351-CF8M SS316 TRIM BC SWING DISC (VA C161)	3"	1	S98	VC	PP06-000002011
12	GATE VALVE FLGD-RF CL150 ASTM A351-CF8M SS316 TRIM STELLITE SEATS BB OS&Y FLEX. WDG DISC HW. OP (VAV151)	3"	1	S98	VA	PP02-000010434
13	GATE VALVE SW*SCR'D (NPT) CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS&Y SOLID WDG DISC HW. OP (VAV005A)	3/4"	1	S98	VAST	PP02-000010451
14	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443
15	FLOW ATTA(FORWARD)	3"	1			-
16	T4-B-01-603	3"	1			T4-B
17	BAG1-S-E30N W30S-240-S	3"	1			BAG-S

INSTRUMENTS

18	TAG NO : PG-9405-940-01, INSTRUMENT	3/4"	1			INSTRUMENT

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	278	3		<2>	266	3	
<3>	100	3		<4>	598	3	

133

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-12-26

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P0AN
P&ID NO.	K1-DSL-UT-PID-9405			PROJ. NO.	16480	FLUID	SPW
CAD NO.	-			LINE NO.	3-SPW-9405002-S98-ISN		
DWG NO.	P0AN-SPW-9405002-01			SHT NO.	1 OF 1	REV.	2

PIPE modified at 17:01:16 Dec-2025, Drawing Created at 13:14:47 19 December 2025
PIPE NAME: 3-SPW-9405002-S98-ISN_P0AN-01

NOTE : THE ORIGINAL AND ALL COPIES OF THIS DRAWING TOGETHER WITH COPYRIGHT ARE THE SOLE PROPERTY OF LEC, SEOUL, KOREA AND SHALL NOT BE USED FOR ANY PURPOSE WITHOUT WRITTEN PERMISSION OF LEC.